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# Sensor Structure Shapes the Format of Cognitive Maps in Navigation Agents

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## Abstract

1 Visual perception and spatial memory in embodied AI are typically treated as  
2 independent modules, but scaling visual encoders often fails to yield robust spa-  
3 tial representations. We investigate whether sensor structure inherently shapes  
4 the format of spatial memory. We train a group of RL navigation agents where  
5 architecture, task, and training are held strictly fixed, varying only the spatial  
6 bandwidth of the visual encoder. We demonstrate that while all agents reach behav-  
7 ioral convergence, their hidden-state position codes fundamentally diverge. Under  
8 a vision-limited (bottleneck) encoder, position condenses into a scene-invariant,  
9 linearly-readable axis. Under a rich encoder, position information scatters into  
10 non-linear, scene-conditional directions. We frame this format shift as an emergent  
11 capacity allocation tradeoff between pre-formatted visual features and recurrent  
12 temporal integration. Furthermore, causal interventions reveal a strict dissociation  
13 between probe-readability and policy reliance, exposing a critical blind spot in cor-  
14 relational interpretability methods. Our findings position sensor visual structure as  
15 a fundamental lever on cognitive-map format, generating architecture-agnostic pre-  
16 dictions for transformer-based navigators, and suggesting that bandwidth-restricting  
17 perception (such as foveation) is actively cognition-enabling rather than merely  
18 compute-saving.

## 19 1 Introduction

20 Visual perception and downstream memory are typically treated as independent modules in a se-  
21 quential pipeline. Under this modular approach, the encoder is developed as a standalone feature  
22 extractor, guided by the assumption that scaling the encoder’s capacity will provide a richer set of  
23 features for the rest of the system. Recent work, however, demonstrates that this “plug-and-play”  
24 logic is flawed: even when equipped with high-capacity encoders that excel at object recognition,  
25 vision–language models (VLMs) consistently fail at tasks requiring an understanding of 3D layouts  
26 and relative positioning [Ramakrishnan et al., 2025, Yang et al., 2025]. While others argue that we  
27 simply need more spatial training data [Chen et al., 2024] or different pretraining objectives [Tong  
28 et al., 2024, El Banani et al., 2024], we propose that the issue lies in the isolation of the components,  
29 as cognitive neuroscience has long recognised that sensor structure shapes the format of cognitive  
30 maps [Wirth et al., 2017, Rolls, 2023, Geva-Sagiv et al., 2016, Fortin et al., 2008]. By treating  
31 perception and memory as a single, co-trained system, we investigate how the architectural structure  
32 of the encoder itself constrains what spatial information can be read from memory.

33 We bridge this gap in a silicon setting. Building on deep-RL navigation agents known to develop  
34 cognitive-map-like representations even in the absence of vision [Wijmans et al., 2023, Banino et al.,

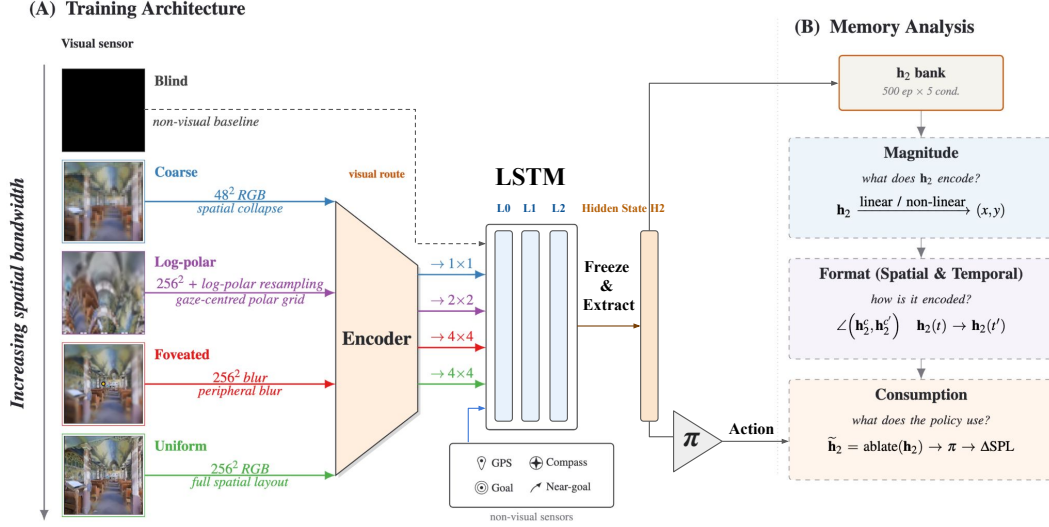


Figure 1: **Experimental setup and analysis pipeline.** (A) Five sensor conditions (leftmost column) ordered by visual bandwidth feed a shared encoder and memory; non-visual sensors enter the memory directly. The top-layer hidden state  $h_2$ , which we read as the agent’s spatial memory, drives the policy. (B) Read spatial memory along three views: Magnitude (decodability, §3), Format (intrinsic structure, §4.1–§4.2), and Consumption (behavioural intervention, §5).

2018, Cueva and Wei, 2018],<sup>1</sup> we introduce a comparative testbed that holds the encoder backbone, memory, task, optimisation, and reward entirely fixed, and varies a single degree of freedom: the visual bandwidth of the visual sensor across five conditions (Figure 1).

We hypothesise an emergent *capacity-allocation* principle: the visual encoder and the integration of continuous sensor history (e.g., GPS) compete for a finite hidden-state budget. Our results are consistent with this tradeoff. As visual bandwidth increases, the linearly-readable, world-frame position signal in memory collapses monotonically, even though every agent still reaches the goal. We characterise this reallocation along three complementary views: *magnitude* (how much linearly-accessible position information lives in memory), *format* (which subspace carries it across conditions), and *consumption* (which route the policy actually uses for behaviour). To guard against single-method artefacts, we triangulate findings across multiple probe families and held-out scenes, and ground the format shift behaviourally with asymmetric memory transplants: rich-encoder policies fail to navigate using bottleneck states, suggesting the representational change is intrinsic to the policy’s use of memory rather than an artefact of the probing framework [Molinari et al., 2025].

Our contributions are three. (i) **The capacity-allocation hypothesis.** A candidate framework relating visual bandwidth to spatial-memory format along three complementary views (magnitude, format, consumption). The matched-condition design opens an analytical avenue biology cannot easily realise: the systems-neuroscience toolkit applied to a controlled comparative population rather than a single recording. (ii) **Probe-readability dissociates from policy mechanism.** Two off-diagonal cases (a bottleneck-encoder agent under-weighting its own readable position code; a rich-encoder agent reading a scene-history substitute despite no readable code) and a complementary causal ablation jointly show that the linearly-readable axis is one window into a distributed population code, not its substrate. (iii) **Predictions for transformer navigators and perception design.** A falsifiable prediction for transformer-based navigators (§6) and an architecture-side reframing of bandwidth-restricting perception (foveation, log-polar) as plausibly cognition-enabling rather than

<sup>1</sup>“Cognitive map” in our paper is operationalised as the linearly-decodable world-frame position code at the policy-readable hidden state. Aspects of the broader concept [Epstein et al., 2017] are engaged by our LOSO scene-invariance test (relational across rooms) and predictive-horizon analysis (SR-style future-state code, Stachenfeld et al. 2017); hierarchical scene clustering and topological reachability are deferred to follow-up.

Table 1: **Navigation performance and spatial probes.** Per condition: SPL, success rate, and mean episode length ( $n=200$  deterministic-rollout episodes, mean  $\pm$  SEM); Ridge probe  $R^2$  for decoding GPS position from  $\mathbf{h}_2$  at linear and 2-layer-MLP probe depths (5-fold episode-level CV mean  $\pm$  std).

| Condition | SPL             | Success | Steps        | GPS $R^2$        |                  |
|-----------|-----------------|---------|--------------|------------------|------------------|
|           |                 |         |              | linear           | MLP-2            |
| Blind     | 0.43 $\pm$ 0.02 | 0.76    | 416 $\pm$ 34 | +0.93 $\pm$ 0.04 | +0.98 $\pm$ 0.00 |
| Coarse    | 0.81 $\pm$ 0.02 | 0.97    | 154 $\pm$ 19 | +0.58 $\pm$ 0.10 | +0.75 $\pm$ 0.09 |
| Foveated  | 0.88 $\pm$ 0.02 | 0.97    | 141 $\pm$ 21 | +0.18 $\pm$ 0.66 | +0.79 $\pm$ 0.05 |
| Log-polar | 0.86 $\pm$ 0.01 | 0.99    | 124 $\pm$ 15 | −0.03 $\pm$ 0.76 | +0.63 $\pm$ 0.17 |
| Uniform   | 0.87 $\pm$ 0.02 | 0.98    | 127 $\pm$ 18 | −1.79 $\pm$ 2.99 | +0.35 $\pm$ 0.80 |

only compute-saving, complementing data-side accounts [Chen et al., 2024] of VLM spatial-reasoning failures.

## 2 Navigation Agents with Sensors of Varying Visual Bandwidth

**Agent configurations** We treat sensor structure as the single degree of freedom (Figure 1A; encoder, memory, sensor stack, training, and reward held identical across conditions). Agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] with DD-PPO [Wijmans et al., 2020]. The core architecture is a 3-layer LSTM (512-d) reading from a shared non-visual sensor stack (goal-vector, GPS, compass, prev-action), with a linear policy head that consumes the top-layer hidden state  $\mathbf{h}_2$ . The bandwidth axis spans two regimes at its endpoints: a *bottleneck regime* (blind, coarse) where the encoder is bypassed or cannot resolve spatial features; and a *rich-encoder regime* (foveated, uniform) where the encoder supplies position-correlated visual features per step. Log-polar is added as a control: log-polar resampling (mirroring the log-radial sampling of primate V1 cortical magnification [Schwartz, 1977]) preserves per-step feature variety while collapsing the encoder feature map, letting us partially disentangle encoder spatial-output dimensionality from per-step feature variety.

**Experimental setup** Each agent is trained from scratch for 250M frames on a merged pool of Gibson 0+ [Xia et al., 2018] (411 scenes) and Matterport3D-train [Chang et al., 2017] (61 scenes); following standard practice in comparative-cognition modelling [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023], we use a single training seed per condition. After training, weights are frozen and per-step  $\mathbf{h}_2 \in \mathbb{R}^{512}$  is collected from  $n=500$  deterministic-rollout episodes per condition on the same scene pool (new (start, goal) pairs sampled fresh, not seen during training). The default decoder is a Ridge linear probe ( $\alpha=10$ , 5-fold episode-level cross-validation, with Hewitt–Liang permutation control [Hewitt and Liang, 2019]). The linear  $R^2$  for decoding GPS position from  $\mathbf{h}_2$  serves as a proxy for how directly the policy’s linear readout can access world-frame position from memory.

**Memory divergence under convergent task performance** Despite different visual inputs, all five agents reliably reach the goal (Table 1; 76–99% Success, SPL 0.43–0.88), with blind less path-efficient than the four sighted as expected of vision-free integration [Wijmans et al., 2023]. To examine how the encoder–memory–policy stack handles this input variation internally, we read  $\mathbf{h}_2$  along three views: magnitude (how much linearly-readable position lives in  $\mathbf{h}_2$ , §3); format (the shape it takes spatially and temporally, §4.1–§4.2); and consumption (which information the policy actually uses under causal intervention, §5).

## 3 Magnitude: Linear Position Code Weakens with Encoder Bandwidth

We hypothesise that the agent can localise via two routes: integrating the GPS-sensor stream over time, or reading a position-correlated feature directly off the encoder. The policy’s readout from  $\mathbf{h}_2$  is linear, so what a linear probe at  $\mathbf{h}_2$  recovers is what the policy can use. Across the bandwidth axis, linear GPS  $R^2$  declines monotonically from +0.93 in blind to −1.79 in uniform (Figure 2a, Table 1) while every agent still reliably reaches the goal, a representational dissociation rather than a skill gap. A low  $R^2$  is consistent with three interpretations: position is absent from  $\mathbf{h}_2$ , distributed non-linearly (§4.1), or replaced by another feature the policy reads (§5).

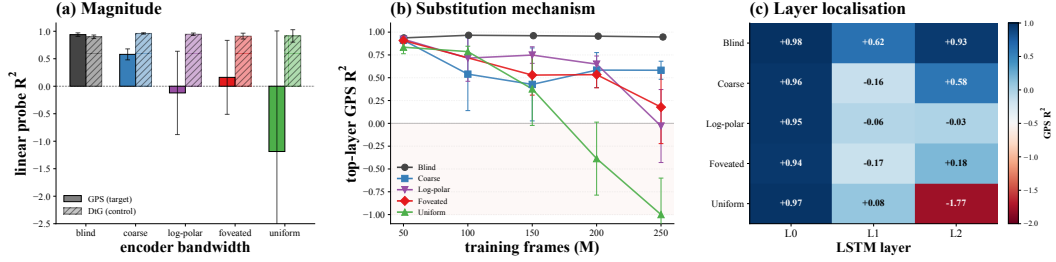


Figure 2: **Memory magnitude.** (a) Per condition, paired bars give linear probe  $R^2$  for GPS (filled) and distance-to-goal (hatched, control). Dashed line at  $R^2=0$ : predict-mean baseline. (b) Top-layer GPS  $R^2$  at five training snapshots. (c) GPS  $R^2$  along the LSTM stack ( $L_0$  input  $\rightarrow L_2$  top, same 5-fold CV protocol as (a, b)).

Before any of those readings, two artefactual explanations need ruling out. (i) Linear probes might just be failing on rich-encoder  $h_2$ . They are not: the same linear decoder reads distance-to-goal at  $R^2 \geq 0.88$  in every condition (Appendix C); the probe has the capacity, so rich-encoder  $h_2$  simply does not carry world-frame position as a linear target. (ii) The bottleneck-rich split might just track encoder output dimensionality. It does not: log-polar’s  $\sim 2 \times 2$  feature map sits close to coarse’s  $1 \times 1$  in raw cell count, yet lands with foveated and uniform on every magnitude diagnostic (Appendix H). The relevant variable is per-step feature variety, not encoder geometry.

With the route-shift reading secured, the next question is when in training the allocation appears. Across training (Figure 2b), all sighted conditions begin at high linear readability (encoder weights are still random, so memory falls back on integration). The split appears as the encoder becomes informative: rich-encoder conditions decay as they transfer position from memory to the visual route, while coarse’s  $1 \times 1$  output cannot carry a position code so its memory plateaus, and blind has no encoder route at all so its memory stays stable. The substitution begins around or after task convergence ( $\sim 100$ M frames for sighted), consistent with a later refinement of an already-working policy rather than co-emerging with skill.

Where in the LSTM the allocation appears (Figure 2c): GPS  $R^2$  at the LSTM input  $L_0$  is uniformly  $\sim 0.95$  across conditions: the GPS sensor enters memory intact regardless of vision.  $L_1$  is dim everywhere; the bandwidth-graded split emerges only at  $L_2$ , the policy-readable layer. The decline in linear decodability is therefore not a failure of sensor integration, but a specific property of the final, policy-accessible representation.

Three independent readings converge: bandwidth-graded magnitude (Fig. 2a), training-time emergence after task convergence (b), and layer-localised divergence at  $L_2$  (c). The encoder’s bandwidth tilts a finite-capacity memory from the integration route toward the visual route, and the divergence shows up specifically where the policy reads. Two further unit-level diagnostics (peaked-unit content and a mid-episode random-action perturbation) converge on the same partition (Appendix D). However, whether the missing linear position is fundamentally discarded, non-linearly redistributed, or substituted by another feature still stays an open question. Sections 4.1 and 5 resolve the critical mechanistic question left open by §3.

## 4 Format: Position is Encoded in Distinct Forms Across Conditions

### 4.1 Spatial Format: Collapsed Linear Readability is a Format Shift, not An Information Loss

The magnitude collapse of §3 does not destroy position information. Non-linear readouts of  $h_2$  recover position where the linear probe fails, and information-theoretic estimates of how much position is encoded shift far less across conditions than the linear  $R^2$  does: the collapse is a format shift, not an information loss. The encoder rearranges position along three axes (manifold geometry, scene-invariance, subspace identity) whose boundaries do not coincide.

The format shift is directly visible in the manifold geometry. A 2D PCA of top-layer hidden states coloured by ground-truth  $x$ -coordinate (Figure 3a) shows blind organising into a single position-graded manifold; coarse is similar but more diffuse; the three rich-encoder conditions fragment into

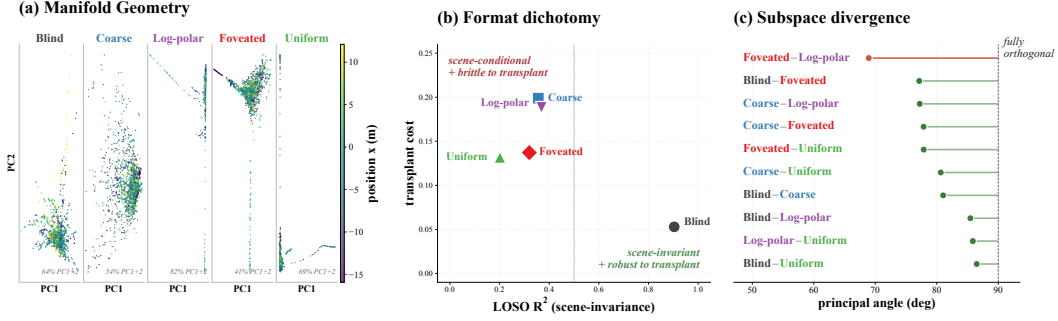


Figure 3: **Memory spatial format.** (a) PCA-2D projection of top-layer  $h_2$  per condition (1500 random steps each), coloured by ground-truth  $x$ -coordinate. (b) LOSO median  $R^2$  for single-axis transfer across rooms vs. cross-condition memory-transplant cost. (c) Principal angle between every pair of conditions’ position-direction subspaces (sorted, with the within-foveated outlier highlighted).

multiple disjoint clouds, each roughly a scene. Inside each cloud the colour gradient is intact, and a 2-layer MLP recovers position at  $R^2 \geq 0.35$  even for uniform (Table 1): the cross-scene linear axis a global probe would need has been replaced by scene-conditional gating.

Scene-invariance follows a strict binary split. We test whether a single spatial axis transfers across held-out rooms (LOSO) and whether one condition’s memory can drive another’s policy (cross-condition transplant cost; Figure 3b). Blind uniquely occupies the scene-invariant and transplant-robust quadrant (LOSO  $R^2 \approx 0.85$ , transplant cost  $\approx 0.05$ ). Any visual input, even the heavily bottlenecked coarse, pushes memory into the scene-conditional and transplant-brittle space. The split sits at the zero-vs-nonzero encoder-bandwidth line, not at low-vs-high: scene-invariance is binary on vision even while magnitude grades continuously.

Subspace identity diverges across all five conditions. Principal angles between position-direction subspaces (Figure 3c) place 9 of 10 cross-condition pairs at  $\geq 75$ ; only the two foveated variants, which share the foveated retina, sit closer at  $\sim 67$ : the five position codes are otherwise nearly orthogonal. Rather than converging on rotated, Procrustes-equivalent variants of a single shared map, identical architectures under different sensors settle into nearly orthogonal memory structures. Sensor structure does not merely modulate a common cognitive map; it appears to select among categorically different ones, with each condition’s  $h_2$  encoded in a format specific to its own policy.

The three boundaries do not coincide, and that is the structural finding: manifold organisation grades from unified (blind) through diffuse (coarse) to fragmented (rich-encoder); scene-invariance flips binary at vision-on/off; subspaces are condition-specific. Visual bandwidth is therefore a multi-dimensional format lever, not a single knob. This recasts §3’s magnitude collapse: rich-encoder conditions hold position in scene-conditional, non-linear directions, so the global linear probe sees nothing even though the information is still there. Mutual-information estimates of  $I(h_2; pos)$  corroborate the preserved-content reading: total position content stays within a factor of  $\sim 1.3$  across conditions even as linear  $R^2$  collapses by orders of magnitude (Appendix D). The blind-vs-sighted dichotomy parallels the no-remapping vs. global-remapping spectrum of hippocampal coding under hidden-state inference [Sanders et al., 2020]. Biologically, this predicts that blind animals rely on a fundamentally different memory format, not a lower-fidelity version of a sighted map.

## 4.2 Temporal Format: Constructed vs Pre-Built Map

Beyond spatial format, visual bandwidth changes how memory operates over time. Vision lets the agent anchor its position in a fixed external frame; without vision, position has to be continuously computed from self-motion, and the internal spatial code updates with every step. Blind sits at the active end, sighted at the stable; within sighted, less encoder bandwidth stiffens the readout. The point is not that blind has a cognitive map (Wijmans [Wijmans et al., 2023] showed it does), but that an integrated motion-based map and a stable vision-based map are categorically different computational objects, not lower-fidelity versions of one shared representation.

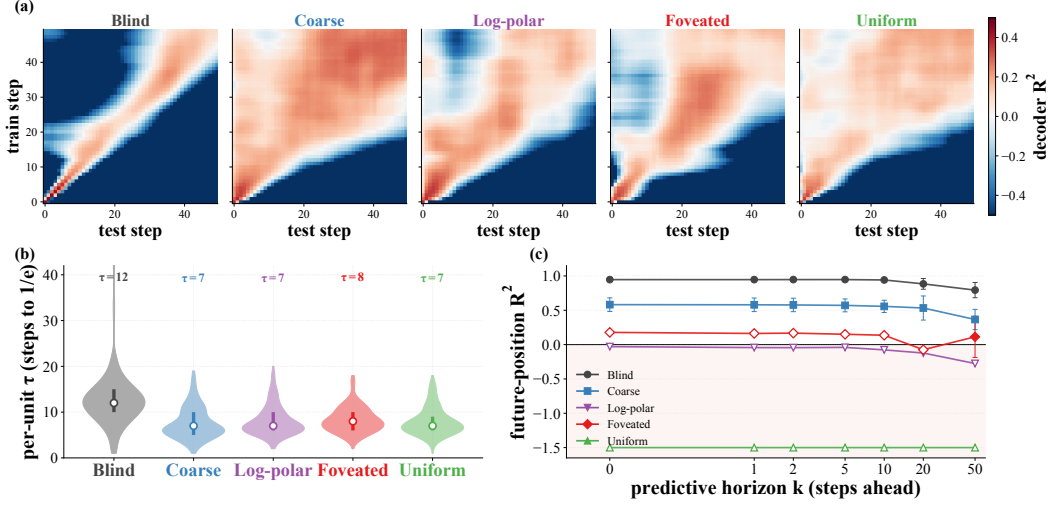


Figure 4: **Memory temporal format.** (a) *Is the readout axis stationary or does it rotate?* A linear decoder of egocentric goal-vector (offset from agent to goal) trained at one trajectory step and tested at every other step. A square block indicates a stationary readout (coarse); a diagonal stripe indicates a step-by-step rotating one (blind). (b) *How long does a memory unit hold onto its past?* Distribution of per-unit intrinsic timescale  $\tau$  across units, defined as the within-episode lag at which the unit’s autocorrelation falls to  $1/e$ . (c) *Does memory carry a forward-rolling trajectory?*  $R^2$  of decoding position  $k$  steps ahead from current memory; shaded band:  $R^2 < 0$ .

To distinguish the two regimes we ask a simple question: *is the memory’s readout axis stable across the episode, or does it rotate with each step?* The natural readout to test is the egocentric goal-vector (the relative offset from the agent’s current position to its goal, expressed in the agent’s own reference frame), because that is the spatial quantity the policy must consume to navigate. A linear decoder of this goal-vector, trained at trajectory step  $t_{\text{train}}$  and tested at every other step  $t_{\text{test}}$  [King and Dehaene, 2014], answers the question directly. A stable, world-anchored map yields a square block in the  $(t_{\text{train}}, t_{\text{test}})$  heatmap (a single decoder transfers across all steps); a step-by-step constructed map yields a diagonal stripe (the decoder works only near the step it was trained at). Blind shows the diagonal; coarse the largest block; log-polar, foveated, and uniform graded between (Figure 4a). A constructed map cannot be stationary because the construction process is the rotation.

Beyond what the heatmap reveals about the readout axis, we ask a more basic dynamical question: *how long does a memory unit hold onto its past activity before decorrelating?* This is the *intrinsic timescale*, a property that separates slow-integrating systems from fast-responding ones. We measure it per unit as the autocorrelation of  $\mathbf{h}_2$  over within-episode steps and read  $\tau$  as the lag at which the autocorrelation crosses  $1/e$  [Murray et al., 2014]; Figure 4b shows the distribution of  $\tau$  across units per condition. Blind has a wide, tall violin (some units integrate over 30+ steps; median  $\tau \approx 12$ ); every sighted condition collapses to a narrow violin at  $\tau \approx 7-8$ . The dichotomy is deeper than the medians: blind’s units are heterogeneous in their timescales (some fast, some very slow) while sighted units are uniformly fast across the population. The two routes segregate timescales: the encoder route ticks fast (every step refreshes from current vision), the integration route runs slow (each step adds a small correction to a persistent state). This mirrors the slow-prefrontal-vs-fast-sensory cortical-hierarchy gradient [Murray et al., 2014] within a single recurrent module, isolating the route effect with anatomy and cell type held constant, a comparison that is hard to set up in biological systems where routes typically map onto distinct brain areas.

The same route segregation predicts a forward-looking signature: if the integration route accumulates a motion trajectory, position should be linearly predictable from memory not just at the current step but several steps ahead. The predictive horizon test (Figure 4c) trains a linear decoder of  $\text{pos}_{t+k}$  from  $\mathbf{h}_t$  at increasing lags  $k$ , the successor-representation cognitive-map signature [Stachenfeld et al., 2017]. Blind sustains  $R^2$  from 0.95 at  $k=0$  to 0.79 at  $k=50$ , a 50-step linear predictive horizon. Every sighted condition decays to chance by  $k=20$  and stays near or below zero from there. The integration route therefore encodes more than current position: it carries a forward-rolling trajectory the linear

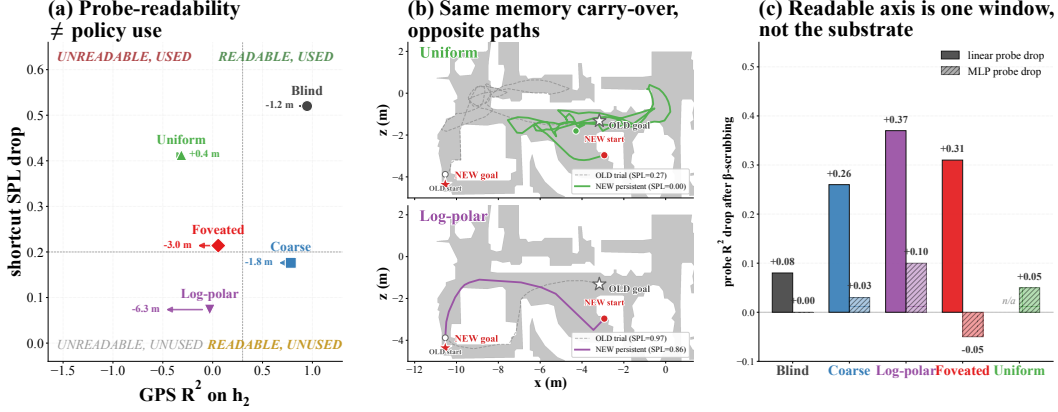


Figure 5: **Probe-readable position is not what drives behaviour.** (a) Each condition’s ( $x$ =GPS  $R^2$ ,  $y$ =shortcut SPL drop) position locates it in the readability–reliance plane. The horizontal arrow encodes the per-condition median paired-episode pull margin  $d(\text{end}, \text{new}) - d(\text{end}, \text{old})$  on same-floor persistent-failure episodes: rightward indicates “locks old”, leftward indicates “tries new”, length is proportional to  $|\text{median}|$ . Only Uniform’s arrow points right. (b) *Two trajectories diverge under identical setup*: Uniform fails to navigate to NEW-goal (its persistent memory pulls it toward OLD-goal area), Log-polar succeeds. Top-down trajectories on one scene’s navmesh for the paired episode with the sharpest contrast in (a). Same scene, same paired (OLD, NEW) episode pair, same memory carry-over from OLD to NEW; only the visual encoder differs across the two panels. Each episode has its own independently-sampled start and goal (so OLD-start  $\neq$  NEW-start, OLD-goal  $\neq$  NEW-goal); the diagnostic question is where the NEW persistent trajectory ends relative to OLD-goal vs. NEW-goal. *Top*: Uniform’s NEW persistent path circles back near OLD-goal (1.2 m from OLD-goal, 6.8 m from NEW-goal). *Bottom*: Log-polar’s path converges on NEW-goal (8.0 m from OLD-goal, 0.0 m from NEW-goal). Dashed grey: OLD trial (faded, for context); solid colour: NEW persistent. (c) Drop in probe  $R^2$  after  $\beta$ -subspace scrubbing. We fit a Ridge  $\beta$  that linearly predicts  $(x, y)$  from  $h_2$ , then project  $h_2$  orthogonally to  $\beta$  at inference. Filled bars: linear probe drop (the ablation collapses the readable axis where it existed). Hatched bars: MLP probe drop (tiny across conditions, so non-linear decoders still recover position from non- $\beta$  directions). Uniform’s linear bar is omitted because its original linear  $R^2$  was already negative noise, so a post-scrubbing change there is uninterpretable.

205 decoder can extrapolate. The encoder route delivers only the current step; without forward-trajectory  
 206 structure, future position becomes unreadable from  $h_t$  alone.

207 The three findings converge on a single structural claim: temporal stability cannot be achieved  
 208 by recurrent capacity alone, because the integration of motion *is* a rotation; stationarity requires  
 209 an external anchor that the encoder provides. Blind reaches the goal not by holding a static map  
 210 but by continuously rebuilding one; the cost shows up as longer paths, but the cognitive map  
 211 itself remains real and accessible (forward-predictable, position-preserving). Persistent-homology  
 212 corroborations [Gardner et al., 2022] are in Appendix F.

## 213 5 Consumption: Probe-Readability Dissociates from Policy Reliance

214 A linear probe reveals what memory stores; it does not verify what the policy actually relies on. To  
 215 test this, we need an intervention that asks the policy a question it can only answer with its memory:  
 216 when memory is helpful, does taking it away hurt? Inspired by Tolman’s same-scene cognitive-map  
 217 test [Tolman, 1948], we run successive eval episodes within each scene under two memory regimes:  
 218 in the persistent regime the LSTM hidden state carries over between episodes, and in the reset  
 219 regime it is zeroed before each. Each episode is sampled independently (its own start, its own goal),  
 220 so consecutive episodes within a scene differ in target. The SPL penalty of persistent relative to  
 221 reset measures how much the policy leans on memory carry-over.

Two off-diagonal cases dissociate readability from reliance (Figure 5a): coarse carries a strong linear position code yet shows little reset penalty (“readable but unused”); uniform carries no linear code yet shows the largest reset penalty (“unused but readable via a substitute”). They raise distinct questions — *what is uniform reading instead of position?* and *is the readable axis where the policy actually reads?* — which two interventions answer.

**What is uniform reading?** When a persistent agent fails the new task, it either drifts toward the old goal (“locks old”) or struggles around the new one (“tries new”). The signed margin  $d(\text{end, new}) - d(\text{end, old})$  (horizontal arrows in Figure 5a) separates the two regimes: only uniform points right (locks old; median +0.43 m); blind, coarse, foveated, and log-polar all point left, increasingly so. Figure 5b shows the contrast on a single scene. What uniform reads looks more like *which scene* than *where in the scene*: a linear scene-id classifier on  $\mathbf{h}_2$  separates scenes far better in sighted than in blind. The simplest reading is a scene-history anchor: “head to where I was going last time I was in this scene”.

**Is the readable axis the substrate?** The coarse off-diagonal raises the converse worry: maybe the linear axis is *epiphenomenal* — correlated with position but not what the policy uses. We ablate it directly: fit a Ridge  $\beta$  predicting  $(x, y)$  from  $\mathbf{h}_2$ , project  $\mathbf{h}_2$  orthogonally to  $\beta$  at inference. The ablation collapses the linear probe but barely moves the MLP probe (Figure 5c;  $\Delta R^2 \in [-0.05, +0.10]$ ): position is distributed redundantly across many directions, with the readable axis one window into a population code rather than its substrate — the redundancy familiar from biological place-cell systems [Stringer et al., 2019]. Read this way, the dissociation in (a) is what we should expect: *readable* and *used* can come apart in either direction.

Two controls affect all conditions equally. Memory-init transplant (re-initialise from the previous-episode end-state rather than zero) drops SPL uniformly: persistent memory is an *activity-slot* store bound to trajectory dynamics [Whittington et al., 2025], not synaptic. GPS+compass ablation from step 0 collapses success to near-zero, confirming GPS as the upstream causal input. Full  $4 \times 4$  catalog in Appendix B.

## 6 Discussion

**Three views of capacity allocation** A reader could imagine three independent things happening across our five conditions: they differ in how much position lives in memory (magnitude, §3), in the form that position takes (format, §4.1–§4.2), and in whether the policy actually uses it (consumption, §5). We read the three as views of one allocation: the visual encoder and the integration of the position sensor compete for finite memory capacity, and visual bandwidth tilts the balance. Bottleneck conditions retain a linear position code in memory; rich-encoder conditions reallocate to the visual route; the policy reads from whichever route the memory makes available. Blind sits alone with a linearly-readable, format-isolated code and the largest behavioural reliance on its memory carry-over; uniform, foveated, and log-polar cluster as non-linear-only, format-shared codes; coarse falls off-diagonal (linearly-readable but format-shared), the configuration that triggers the dissociation in §5. Coarse and uniform end up on opposite sides of the magnitude axis for different reasons: coarse’s encoder cannot supply spatially-resolved features, so capacity defaults to the integration route; uniform’s encoder supplies rich features that capture per-step position, so capacity is reallocated to the visual route, leaving integration unattended. The two are different regimes of one allocation, not the same mechanism.

**Biological precedent** The “sensor format shapes spatial memory” hypothesis has biological precedent [Chen et al., 2016, Geva-Sagiv et al., 2016, Fortin et al., 2008]: subterranean rodents that rely on integration rather than vision [Kimchi et al., 2004] parallel our blind agent; uniform’s near-isotropic sampling parallels insect and zebrafish vision [Geva-Sagiv et al., 2015, Toledo et al., 2020]; coarse’s heavily-bottlenecked input parallels biological systems restricted to low-resolution environmental cues; and the cortical magnification structure of mammalian visual systems [Strasburger et al., 2011, Rosenholtz, 2016] parallels our log-polar control. Our results also align with the hidden-state inference framework of Sanders et al. [2020]: rich-encoder agents fail to generalise position codes across rooms (the LOSO result of §4.1), creating scene-specific hidden states that mimic biological global remapping, while bottleneck agents show no remapping. We do not claim direct biology–AI equivalence; we offer the parallel as a hypothesis the comparative-cognition literature can test

by within-species manipulations of effective per-step visual variety (gaze restriction, dark-rearing, prosthetic compound vision).

**Implications and scope** The principle is geometric: finite memory has fewer effective dimensions than its parameter count, and what those dimensions encode reflects a content-level competition between sensor sources. When the encoder pre-formats per-step features that already include position, memory reallocates capacity away from integration, eliminating the linearly-readable position code; under a bottleneck encoder, integration dominates and the linear code returns. This matches the Information Bottleneck invariance–disentanglement prediction [Tishby et al., 1999, Achille and Soatto, 2018]: bandwidth-restricted memory drives representations toward scene-invariant rather than scene-conditional encoding (Figure 3b). The principle should also be architecture-level: any system where finite-capacity memory reads a pretrained encoder should show the same competition, regardless of whether “memory” is recurrent state or attention over history. In transformer-based navigators (e.g. Zeng et al. 2024; cf. Whittington et al. 2022), the principle predicts attention-readable position codes under bottleneck encoders and attention-internal, only non-linearly decodable codes under rich encoders, a direct falsification target. The same logic re-frames recent VLM spatial-reasoning failures [Ramakrishnan et al., 2025]: rich pretrained encoders may divert capacity away from the downstream substrate’s spatial code rather than merely fail to teach it, so bandwidth-restricting perception may be *cognition-enabling* rather than only compute-saving. As a partial architecture-independence test, we ported the design to a different environment, encoder family, and training regime: frozen DINOv2-Base [Oquab et al., 2024] patch features at five resolutions feed a small recurrent memory trained on Memory Maze [Pasukonis et al., 2023] with next-feature prediction. The same dichotomy reproduces: the non-linear gap ( $\text{MLP } R^2$  minus linear  $R^2$ ) is monotone-decreasing with bandwidth (coarse +0.62 vs. uniform +0.55; Appendix I), in a setup that omits Habitat’s GPS/compass channel and so isolates the format-shift dynamic from the bandwidth-allocation tradeoff.

**Limitations** We evaluate a single training seed per condition (a convention from comparative-cognition modelling [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018, Schaeffer et al., 2023]). Convergent multi-method evidence (geometrical, behavioural, and informational metrics ranking conditions identically, plus an independent blind replication; Appendix A) argues against single-seed noise as the explanation, but multi-seed confidence intervals remain necessary future work. The pipeline (LSTM + DD-PPO + PointGoal + ResNet-18) matches prior cognitive-modelling work for direct comparability [Wijmans et al., 2023, Banino et al., 2018, Cueva and Wei, 2018]. Whether the magnitude and format patterns generalise to transformer-based navigators, alternative RL algorithms, or other tasks (e.g. ObjectGoal) is the cleanest falsification target. The Memory-Maze partial test (Appendix I; different encoder family, environment, and training regime) replicates the format-shift dichotomy, arguing against a pipeline-specific origin (LSTM gating, DD-PPO, or our particular sensor stack); and cross-condition memory transplants (§5) demonstrate the ordering behaviourally, independent of any probe methodology.

## 7 Conclusion

We demonstrate that an agent’s cognitive map format is not a static architectural given, but an emergent property shaped by its visual sensor’s bandwidth. Under a finite memory budget, rich pre-formatted visual features preempt temporal integration, driving memory into a scene-dependent format. Conversely, restricting sensory bandwidth frees capacity to construct globally linear, integration-derived spatial maps. For embodied AI, this yields a counterintuitive design principle: perceptual bottlenecks are not merely compute-saving compromises, but actively cognition-enabling constraints. Methodologically, our results expose a critical dissociation in mechanistic interpretability: the linear decodability of a representation does not guarantee its use by the downstream policy, underscoring the necessity of causal, comparative-population interventions. Ultimately, our capacity-allocation principle bridges AI and neuroscience, suggesting that biological dichotomies (e.g., place vs. view cells) emerge from distinct sensory ecologies rather than divergent architectures. This opens a new pathway to scale spatial reasoning in foundation models not just by adding data, but by strategically constraining perception.

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## A Reproducibility: Hyperparameters and Implementation Notes

**Training hyperparameters (DD-PPO, all five conditions identical)** 8-GPU torchrun on a single 8×A100 node; num\_steps=256, learning rate lr=2.5e-4 with lr\_decay=False; num\_environments=16/worker; total budget 250M frames per condition. All other hyperparameters follow the Habitat-baselines DD-PPO defaults from Wijmans et al. [2020].

**Architectural details (sensor encoding)** Each non-visual sensor (goal-in-start-frame, GPS, compass, close-to-goal scalar) is projected to 32-d via a per-sensor linear layer; the previous-action embedding is also 32-d (nn.Embedding with action-space cardinality + 1 for the no-prev-action token). The four sensor projections are concatenated with the previous-action embedding and the (flattened) visual encoder output to form the LSTM Layer-0 input vector. This matches the Wijmans et al. [2020] Wijmans-default sensor stack.

**Implementation notes (foveated condition)** The foveated condition disables the running-mean-and-variance input normaliser (a per-channel running normalisation layer applied to the visual stream before the ResNet-18 backbone) to avoid an in-place autograd conflict between the running-buffer update and the foveation-blur backward pass. Behavioural success and SPL are unaffected by this choice; an ablation that re-enables the normaliser (with the autograd conflict patched) is available for future work that wants to revisit it.

**Implementation notes (numerical stability)** Two safeguards protect against rare numerical instabilities arising from off-navmesh episodes: (i) a pre-update gradient sanitisation pass replaces NaN/Inf gradients with zero before each PPO step; (ii) a forward-looking geodesic-distance clamp prevents reward NaNs when the agent steps off the navmesh. Both are no-ops on healthy trajectories and apply identically to all five conditions.

**Log-polar foveation transform** The log-polar control (Appendix H) applies a differentiable log-polar resampling step before the ResNet-18 backbone: the  $128 \times 128$  avg-pooled image is mapped onto a  $64 \times 64$  grid with  $n_\rho=64$  log-spaced radial samples and  $n_\theta=64$  angular samples, mimicking the cortical magnification of mammalian primary visual cortex. Visual front-end only; LSTM, sensors, and training pipeline are identical to the foveated condition.

## B Probing Protocol Details

**Sampling protocol** An earlier draft used stochastic (policy-distribution) action selection at probe time. Conditions with higher action-entropy under their trained stochastic policies sampled the STOP action early with  $\sim 0.25$  per-step probability, producing  $\sim 4$ -step rollouts and target variance two orders of magnitude below the episodic range. The resulting probe  $R^2$  values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons; we switched to the deterministic (argmax) protocol used here and elsewhere in our pipeline. All paper results use the deterministic protocol.

**Length-matching ablation** Under deterministic rollouts, episode lengths span  $\sim 100$ – $400$  mean steps depending on condition. We use 5-fold episode-level CV to keep splits balanced per scene rather than length-matching across conditions. A controlled length-matching ablation (truncate every condition to a common length) is left for follow-up; we expect the magnitude ordering to be preserved because bottleneck conditions decode at high  $R^2$  even at the smallest sub-sample, but have not verified.

**Cross-heading probe** The proposal included a cross-heading generalisation probe (train on heading  $A$ , test on opposite heading). Our implementation returned “insufficient samples” sentinels at our probe sizes; not in our results.

**LSTM Layer 0 sensor branch** The non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar, prev-action) projects each sensor to 32-d, concatenates them with the visual encoder’s flattened output, and feeds the result as the LSTM’s Layer 0 input. The GPS embedding therefore enters the LSTM via this concatenation; this explains why Figure 2c shows linearly-decodable GPS at Layer 0 even for conditions whose visual encoder discards GPS entirely. Per-layer probes show the same pattern at every hidden-state layer: Layer 0 carries GPS for every condition; Layer 1 dips for everyone; Layer 2 separates blind/coarse from foveated/uniform. Cell states  $\mathbf{c}_\ell$  do not carry the GPS code as cleanly as hidden states ( $\mathbf{c}_1$  at chance/negative for every condition;  $\mathbf{c}_2$

positive only for blind,  $R^2 \approx +0.57$ ). The headline “top-layer linearly-decodable GPS” claim is therefore specifically about  $\mathbf{h}_2$ , not the cell pathway.

**Non-linear (MLP) probe details** For each condition we fit a 2-layer MLP (hidden 256–128, ReLU,  $L_2$  weight decay  $10^{-4}$ , sklearn MLPRegressor with early-stopping) on the same top-layer hidden states and GPS labels as the linear Ridge probe, with the same 5-fold episode-level CV. The full probe-depth sweep (linear  $\rightarrow$  MLP-1  $\rightarrow$  MLP-2  $\rightarrow$  MLP-4) shown in Figure 3a establishes a depth-graded recovery: bottleneck conditions plateau near  $R^2 \approx 0.95$  from depth-0 (linear suffices), while rich-encoder conditions ramp up sharply with probe depth. Top-layer MLP-2 recovery (Ridge linear  $\rightarrow$  MLP-2): blind  $0.92 \rightarrow 0.98$ , coarse  $0.51 \rightarrow 0.75$ , foveated  $0.06 \rightarrow 0.79$ , log-polar  $-0.07 \rightarrow 0.63$ , uniform  $-2.28 \rightarrow 0.35$ . The MLP probe alone cannot disentangle “non-linear position code” from “MLP regressing through position-correlated features (wall distance, scene identifiers, episode-phase markers)”; the log-polar foveation control (Appendix H) targets that question at one intermediate point.

## C Alternative-Account Explanations

*Task-competence differences.* All sighted conditions solve PointGoal at 93–99% success and blind at 76% (Table 1); all five conditions converge behaviourally well before the probing window, so the encoded-content gap cannot be attributed to navigation skill or differential under-training.

*Probe-capacity / probe-fitting artefacts.* A direct test: distance-to-goal (DtG) decodes robustly across all five conditions ( $R^2 \geq 0.88$ ; Figure 9) while the goal-direction component is at chance everywhere (PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it). The linear probe is therefore capable of fitting position-correlated structure when it exists; the chance-level rich-encoder GPS  $R^2$  is a hidden-state property, not a probe artefact. The real-vs-permuted-label gap (Hewitt–Liang selectivity) likewise tracks GPS  $R^2$ .

*Sample-size / trajectory-distribution artefacts.* CV  $\sigma$  spans come from the same 500-episode pool with  $\sim 100$ –400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft is documented and retired in Appendix B.

*Linear-probe artefacts.* Three complementary checks. (a) *Behavioural*: the memory-transplant intervention goes outside the probe framework and reproduces the bottleneck-vs-rich-encoder ordering; the  $5 \times 5$  matrix’s asymmetry adds structure no linear-similarity measure can read. (b) *Non-linear (MLP) probe*: a 2-layer MLP recovers  $R^2 \in [0.35, 0.79]$  from rich-encoder top-layer states (Appendix B). (c) *Non-linear similarity*. Beyond linear similarity (off-diagonal CKA [Kornblith et al., 2019]  $< 10^{-4}$ ), a non-linear t-SNE embedding also separates the five conditions with 1-NN purity 1.000 vs. chance 0.20 — the format divergence is not an artefact of linear-similarity measures. H1 is therefore specifically about *linear* top-layer readability — what the linear DD-PPO action head can consume — not GPS absence.

## D Magnitude: Corroborating Diagnostics

To anchor the information-allocation finding in §3 on a nat-scale rather than the bounded, model-dependent  $R^2$  scale of probes, we estimate  $I(\mathbf{h}_2; \text{pos})$  per condition using the Mutual Information Neural Estimator [Belghazi et al., 2018]. We fit a 3-layer MLP critic (256 hidden, ELU, batch 512, lr  $10^{-4}$ , 4,000 steps) on joint  $(\mathbf{h}_2, \text{pos})$  samples vs. within-batch shuffled marginals, with the bias-corrected gradient (eq. 12 of Belghazi et al. 2018). Final  $I(\mathbf{h}_2; \text{pos})$  in bits (mean  $\pm$  std across 3 seeds): blind  $6.01 \pm 0.06$ , coarse  $4.55 \pm 0.03$ , log-polar  $4.66 \pm 0.04$ , foveated  $4.30 \pm 0.04$ , uniform  $4.47 \pm 0.06$ . Total mutual information decreases with encoder bandwidth, but the decrease is gentle ( $\approx 1.3 \times$  blind/uniform) compared to the linear-probe  $R^2$  collapse (effectively factor-of-zero for uniform). The within-sighted ordering is reproducible across seeds (per-condition std  $\leq 0.06$  bits, much smaller than the within-sighted gap  $\approx 0.36$ ); the dominant signal is the bottleneck-vs-rich-encoder gap (blind 6.01 vs sighted mean 4.49). Interpretation: rich-encoder conditions retain substantial position information in  $\mathbf{h}_2$ , but it is encoded non-linearly and partly offloaded to the encoder route, paralleling the LOSO scene-invariance result (Figure 3b).

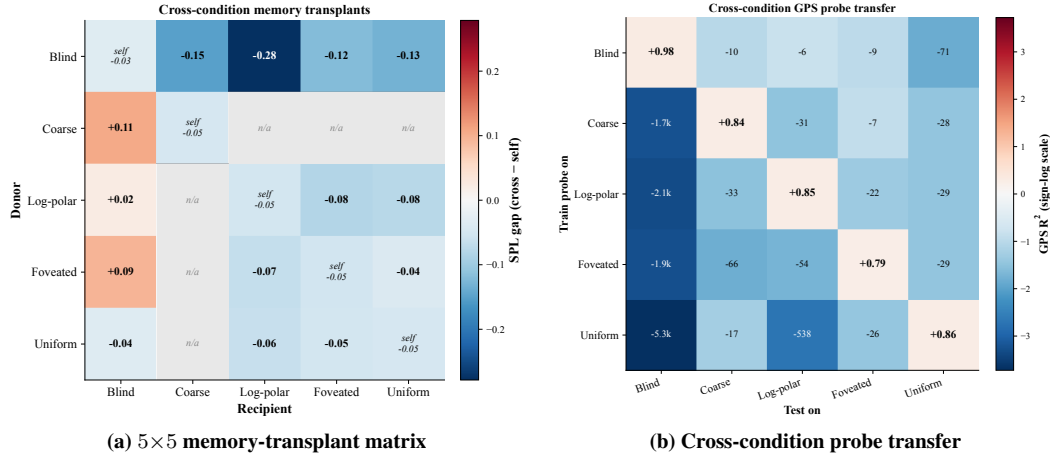


Figure 6: **Two corroborating views of format divergence.** (a) Full 5×5 memory-transplant matrix at midpoint  $t=30$ . Per-recipient column means enter main Figure 3b as the transplant-cost axis; the matrix exposes the underlying direction-asymmetry. (b) Cross-condition probe-transfer matrix: train a Ridge GPS probe on the row condition, test on the column. Off-diagonal  $R^2$  is strongly negative (clipped to  $[-1, 1]$  for display; raw values  $-3.5 \times 10^2$  to  $-3.5 \times 10^4$ ): position-axes are jointly incompatible at the readout level even though every condition visits the same scenes and solves the same task.

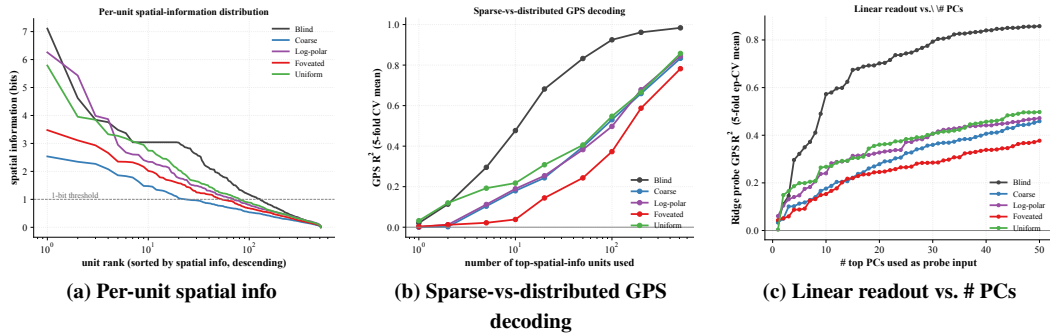


Figure 7: **Capacity-allocation geometry within  $h_2$ .** Three views of the same population structure: bottleneck conditions concentrate the position code on a few axes, rich-encoder conditions distribute it across many. (a) Per-unit spatial-information distribution (units sorted descending; log  $x$ ). Rich-encoder conditions hold a few units with marginally higher peak spatial info ( $\approx 2$  bits) than bottleneck conditions. (b) GPS decodability from the top- $k$  spatial-info units: blind reaches  $R^2 \approx 1.0$  at small  $k$ ; rich-encoder networks fail at every  $k$ . Higher per-unit spatial information  $\neq$  population position readout: peaked rich-encoder units encode features correlated with position (visual landmarks, trajectory phase), not position itself. (c) Ridge-probe GPS  $R^2$  versus the number of top PCs of  $h_2$  used as input: blind plateaus by  $\leq 10$  PCs (axis aligned with low-variance directions); rich-encoder  $R^2$  barely climbs across 50 PCs. The participation ratio  $(\sum \lambda)^2 / \sum \lambda^2$  varies across sighted conditions (1.7–9.1) without tracking encoder bandwidth monotonically.

## 919 E Spatial Format: Corroborating Diagnostics

920 This appendix collects spatial-format diagnostics in four consolidated figures: convergent format-  
 921 divergence metrics (Figure 6); capacity-allocation geometry within  $h_2$  (Figure 7); eigenspectrum  
 922 (Figure 8); and the goal-vector probe (Figure 9). The  $\beta$ -subspace scrubbing intervention is reported  
 923 in main-text Figure 5c. All panels report the full 5-condition set.

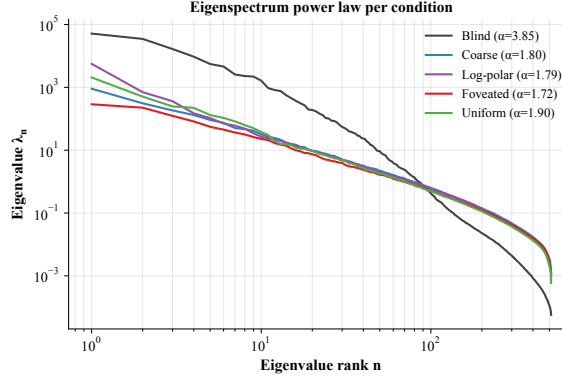


Figure 8: **Eigenspectrum of  $h_2$ : geometric reading of capacity allocation.** Log-log eigenvalue  $\lambda_n$  vs. PC index  $n$ , with reference exponents  $\alpha=1$  (Stringer V1) and  $\alpha=1 + 2/3$  (smoothness/efficiency border at  $d_{\text{task}}=3$ ); per-condition  $\alpha$  fitted on  $\log-\lambda_n$  vs.  $\log-n$ , PCs 5–200. Blind sits well above the critical exponent ( $\alpha \approx 3.85$ ; top-10 PCs hold 95% of variance), a few-direction concentrated regime; the four sighted conditions cluster around  $\alpha \approx 1.7$ – $1.9$ , spreading variance across more directions. The split mirrors the magnitude split (§3) at the geometric level: blind’s reliance on the integration route yields a low-dimensional spectrum; sighted conditions spread variance to support encoder-route information.

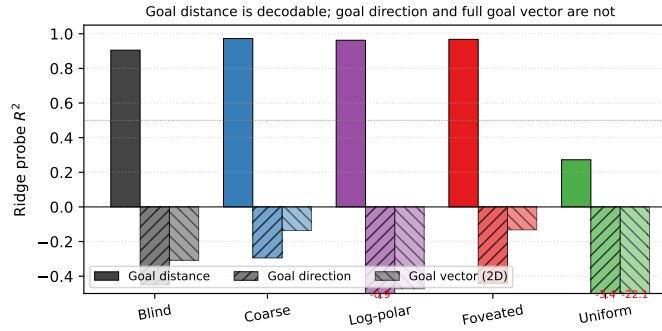


Figure 9: **Goal-vector probe  $R^2$  across all five conditions.** Goal distance (DtG) decodes strongly everywhere ( $R^2 \in [0.89, 0.96]$ ; 5-fold episode-level CV); goal direction and the full ego-frame goal vector are at chance or negative for every condition. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it. The probe is therefore capable of reading position-correlated structure when it exists; the chance-level rich-encoder GPS  $R^2$  is a hidden-state property, not a probe artefact.

## 924 F Temporal Format: Corroborating Diagnostics

925 This appendix collects two further diagnostics referenced from §4.2: a topological-distinguishability  
 926 test via persistent homology (Section F.1) and a predictive-horizon test grounded in successor-  
 927 representation theory (Section F.2). One note on the main-text intrinsic-timescale view (Figure 4b):  
 928 within sighted, the four conditions cluster near-identically at the per-unit autocorrelation level ( $\tau \approx 7$ –  
 929 8 vs. blind’s  $\tau \approx 12$ ); the within-sighted bandwidth-inverse stiffness signal lives in the decoder-decay  
 930 view (Figure 4c) rather than in per-unit timescale.

### 931 F.1 Topological Distinguishability via Persistent Homology

932 A parameter-free pipeline used for toroidal grid-cell structure in mouse cortex [Gardner et al.,  
 933 2022] confirms that conditions are topologically distinguishable. Persistent  $H_1$  loop counts order  
 934 monotonically with encoder bandwidth: blind 219, coarse 203, log-polar 183, foveated 152, uniform  
 935 133. Cross-condition Wasserstein-2 distance between  $H_1$  persistence diagrams ( $\sim 15$ – $20$  dgm<sub>1</sub>) is

roughly  $2\times$  the within-condition seed noise ( $\sim 8\text{--}11 \text{ dgm}_1$ ) for every condition, so the  $\mathbf{h}_t$  manifolds are topologically distinguishable. We treat this as suggestive corroboration rather than independent mechanistic evidence. Our pre-registered prediction was the opposite direction (we expected uniform highest, blind lowest); we report what we found.

## F.2 Predictive Horizon: Does $\mathbf{h}_t$ Encode Future Positions?

Successor-representation theory [Stachenfeld et al., 2017, Whittington et al., 2020] predicts that hippocampal place cells encode not just current position but a discounted future-occupancy code: place-cell firing at  $s_t$  corresponds to expected visits to nearby states  $s_{t+k}$ . We test the analog claim for our recurrent state: does a linear probe from  $\mathbf{h}_t$  recover *future* positions  $(x, y)_{t+k}$  for  $k \in \{0, 1, 5, 10, 20, 50\}$ ?

|                                                                                                               | $k = 0$ | $k = 1$ | $k = 5$ | $k = 10$ | $k = 20$ | $k = 50$ |
|---------------------------------------------------------------------------------------------------------------|---------|---------|---------|----------|----------|----------|
| <i>Linear (Ridge <math>\alpha = 10</math>) <math>R^2</math>, all-episode pooling, episode-level 5-fold CV</i> |         |         |         |          |          |          |
| Blind                                                                                                         | +0.94   | +0.94   | +0.94   | +0.92    | +0.90    | +0.79    |
| Coarse                                                                                                        | +0.57   | +0.57   | +0.57   | +0.56    | +0.50    | +0.37    |
| Foveated                                                                                                      | +0.06   | +0.06   | +0.08   | +0.13    | +0.19    | +0.06    |
| Log-polar                                                                                                     | +0.17   | +0.18   | +0.17   | +0.11    | −0.14    | −0.50    |
| Uniform                                                                                                       | −4.31   | −4.32   | −4.32   | −4.26    | −4.39    | −10.14   |

Blind exhibits a long predictive horizon: linear  $R^2$  from  $\mathbf{h}_t$  to  $(x, y)_{t+k}$  remains  $\geq 0.90$  from  $k = 0$  to  $k = 20$  (only 0.04-point drop) and 0.79 at  $k = 50$  — the same linear projection that decodes current position at  $R^2 = 0.94$  decodes positions 20 steps in the future at  $R^2 = 0.90$ , a strong form of the predictive-map signature predicted by SR theory. Coarse follows at lower magnitude ( $R^2 \in [0.50, 0.57]$  across  $k = 0\text{--}20$ , 0.37 at  $k = 50$ ). Rich-encoder conditions have no clear predictive horizon: foveated stays in  $[0.06, 0.19]$  across all  $k$  with no monotonic decay; log-polar starts at +0.17 and turns negative by  $k = 20$ ; uniform is already strongly negative at  $k = 0$  and grows more negative. Rich-encoder  $\mathbf{h}_t$  encodes neither current nor future position linearly — the LOSO scene-invariance reading of §4.1 extended in time. The asymmetry: bottleneck conditions encode  $\mathbf{h}_t$  as a smoothly-varying integrated GPS code that linearly predicts upcoming positions; rich-encoder conditions encode  $\mathbf{h}_t$  reactively to the current visual scene without temporal extrapolation.

## G Consumption: Corroborating Diagnostics

The main-text §5 reports the persistent-memory failure-mode test as the condition-discriminating intervention; this appendix presents two boundary-check interventions that characterise the consumption mechanism without separating conditions (Section G.1), the full  $5\times 4$  failure-mode catalog spanning the per-condition margin distribution (Figure 10), and a memory-transplant midpoint sweep that defends the  $t=30$  choice in the main-text  $5\times 5$  transplant matrix (Figure 11).

### G.1 Boundary Checks: Memory-Init Transplant and GPS-sensor Ablation

*Memory-init transplant* (re-initialise memory from the previous-episode end-state instead of from zero) drops SPL by a comparable amount across all five conditions ( $\Delta \text{SPL} \in [-0.10, -0.14]$ ). The cross-condition uniformity is itself the finding: every policy at  $t=0$  expects a zero memory state, so memory in this architecture is an *activity-slot* store [Whittington et al., 2025] bound to the trajectory dynamics, not a synaptic store. *GPS+compass ablation* from step 0 collapses success from baseline (0.81–0.98) to near zero (0.02–0.06) across all five conditions ( $\Delta \in [-0.75, -0.97]$ ): the position sensor is the upstream causal source whose integration memory encodes. The absolute drops cluster tightly across the four sighted conditions and do not cleanly separate them, consistent with the upstream sensor being a shared causal bottleneck rather than a discriminator.

### G.2 Persistent-Memory Failure-Mode Catalog

The main-text Figure 5 shows one canonical paired-episode case per condition; the full  $5\times 4$  catalog below shows four cases per condition (now including log-polar) spanning the per-condition margin

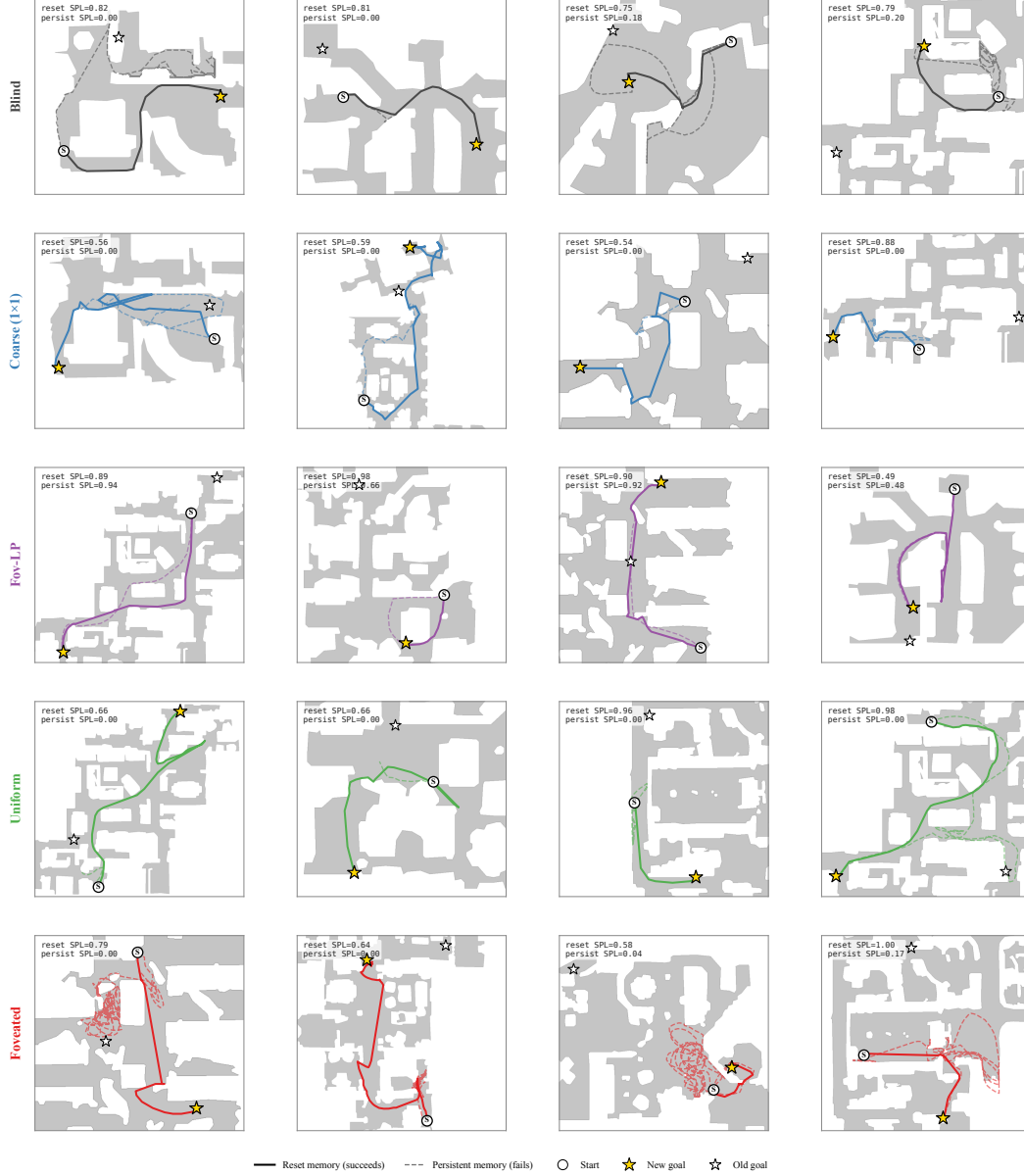


Figure 10: **Persistent-memory failure-mode catalog: 4 cases per condition spanning the per-condition margin range.** Reset trajectory (solid, condition colour) reaches the new goal; persistent trajectory (dashed bold) shows where the persistent-memory agent ends up. Blind and Coarse cluster at “tries-new”; Log-polar’s persistent agent rarely fails outright, so all four selected cases skew tries-new; Uniform clusters at “locks onto old”; Foveated is mixed.

range. Selection criterion: reset SPL > 0.5, persistent SPL < 0.2, persistent steps  $\geq 30$ ,  $|\Delta y_{\text{goal}}| < 0.5$  m (same-floor old vs. new goal). Panels within each row are ordered by margin (left = most-negative = closer to new goal; right = most-positive = closer to old goal).

### G.3 Memory-Transplant Midpoint Sweep: Defending the $t=30$ Choice

The main-text  $5 \times 5$  memory-transplant matrix is reported at a fixed midpoint  $t=30$  (the full matrix appears as panel (a) of Figure 6). A natural concern is whether the asymmetry pattern is specific to that midpoint. The midpoint sweep below addresses this: across midpoints  $\{0, 15, 30, 60, 90, 200, 400, 800\}$  time-steps, the bottleneck-vs-rich-encoder gap appears for  $t \in$

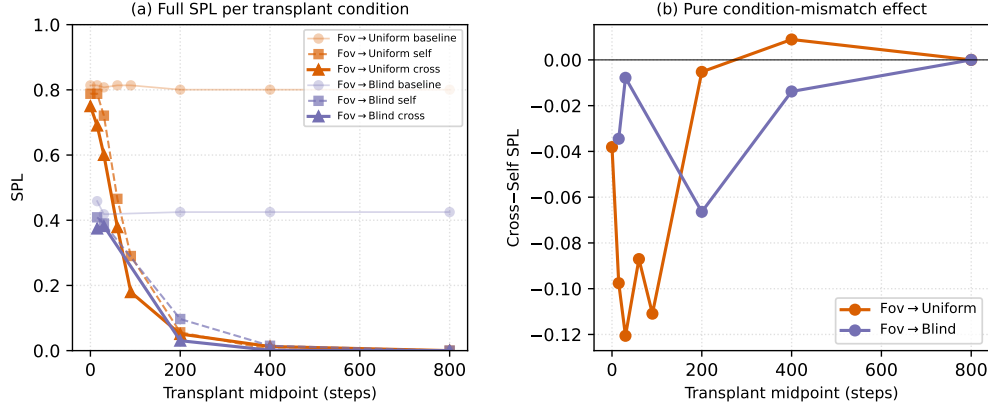


Figure 11: **Memory-transplant midpoint sweep: the  $t=30$  asymmetry is not a midpoint artefact.** Two representative donor→recipient pairs (foveated→uniform, foveated→blind) across midpoints  $\{0, 15, 30, 60, 90, 200, 400, 800\}$  time-steps ( $n=100\text{--}150$  episodes/cell); the main-text  $5\times 5$  matrix is the  $t=30$  slice. **(a)** Full SPL (baseline / self / cross):  $t=0$  collapses to a protocol-noise floor (so the main-text diagonal reflects rollout divergence, not condition mismatch); for  $t \geq 200$  all curves decay because most successful episodes have already terminated. **(b)** Cross–self isolates the pure condition-mismatch effect: it peaks at  $t \in [30, 90]$  (the agent has integrated history but is not yet locked into a trajectory) and decays toward 0 by  $t \geq 400$ .

985 [30, 90] and decays toward zero by  $t \geq 400$ , with a protocol-noise floor at  $t=0$  where the donor’s  
 986 hidden state has had no time to integrate.

## 987 H Encoder-Capacity Scaling: Log-Polar Foveation Control

988 A direct test of the H1 mechanism (encoder spatial-feature variety per step drives top-layer LSTM  
 989 spatial encoding) is to vary that variable continuously across our existing 4-condition spectrum. The  
 990 most diagnostic single point is the *log-polar foveation* control: log-polar resampling onto a  $64\times 64$   
 991 retinal grid produces an encoder spatial output of  $\sim 2\times 2$ , intermediate between coarse’s  $1\times 1$  and  
 992 uniform’s  $4\times 4$  (verified by direct forward pass through the trained encoder). Only the visual front-end  
 993 differs from the foveated condition; LSTM, sensors, and training pipeline are identical. **Preregistered**  
 994 **prediction:** if the H1 mechanism is the simple “encoder spatial-output dimensionality” reading,  
 995 log-polar should produce LSTM GPS  $R^2$  in the bottleneck regime (i.e. closer to coarse’s  $+0.58$  than  
 996 to uniform’s  $\approx 0$ ); if log-polar matches uniform on H1 instead ( $R^2 \approx 0$ ), the encoder-spatial-output  
 997 mechanism story needs reframing toward channel information, encoder spatial-feature *variety* per  
 998 step, or another aspect. **Result (converged 250M frames).** The log-polar agent at convergence (ckpt-  
 999 49,  $\sim 245$ M frames) yields top-layer GPS  $R^2 = -0.029 \pm 0.759$  (5-fold CV,  $n = 500$  episodes) —  
 1000 *not* in the bottleneck regime, but indistinguishable from foveated ( $R^2 = +0.178 \pm 0.659$ ) and clearly  
 1001 far from coarse ( $R^2 = +0.582 \pm 0.099$ ). The simple “encoder spatial-output dimensionality” framing  
 1002 is therefore *not* sufficient to explain the H1 phenomenon: log-polar’s  $2\times 2$  output does not place it  
 1003 in the bottleneck regime. We read the result as supporting the alternative “encoder spatial-feature  
 1004 *variety* per step” framing: log-polar’s radial sampling preserves rich gaze-centred feature variety per  
 1005 timestep even at low spatial output, so the policy can still source position from the visual route and the  
 1006 LSTM does not commit capacity to the integrated code. Substitution-dynamics analysis (Figure 2b)  
 1007 corroborates this reading: log-polar starts at  $R^2 = +0.92$  at  $\sim 50$ M frames (matching all 4 main  
 1008 conditions’ early high  $R^2$ ), then decays along the rich-encoder trajectory rather than plateauing in the  
 1009 bottleneck range. A finer multi-point input-resolution sweep ( $K \in \{32, 64, 96, 128, 192\}$ ) is reported  
 1010 as future work; its training cost (multiple from-scratch DD-PPO runs at  $\sim 250$ M frames each) was  
 1011 outside this submission’s compute window.

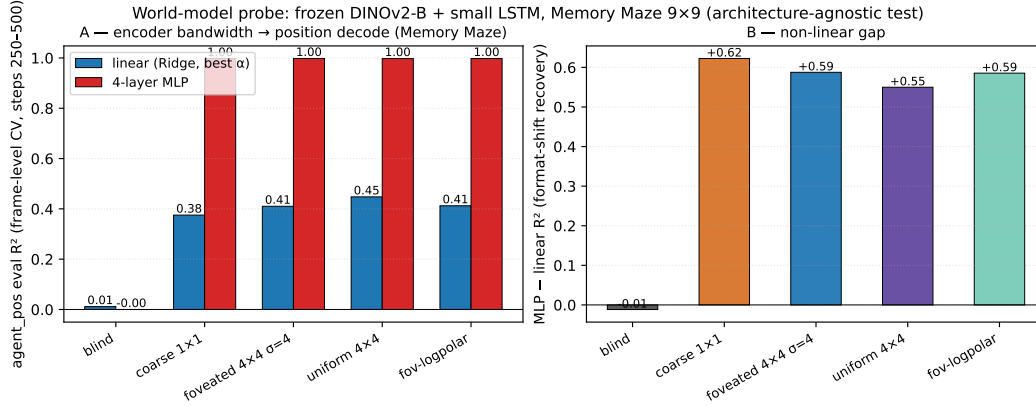


Figure 12: **Format shift reproduces on Memory Maze with a frozen DINOv2-Base encoder.** (a) Linear (Ridge, blue) and 4-layer MLP (red) eval  $R^2$  for agent\_pos from the LSTM hidden state. Linear  $R^2$  rises with encoder bandwidth (blind 0.01 → uniform 0.45); MLP  $R^2$  saturates at  $\approx 0.998$  for all sighted conditions and stays at chance for blind. (b) Non-linear gap (MLP–linear) decreases with bandwidth (coarse +0.62 vs. uniform +0.55): the same format-shift dichotomy as Habitat. The reversed linear- $R^2$  direction across the bandwidth axis reflects Memory Maze’s lack of GPS/compass channels: the encoder is the only sensory route to absolute position, so blind carries no signal at all.

## 1012 I Architecture-Agnostic Confirmation: Memory Maze with DINOv2

1013 Companion figure to the architecture-agnostic claim in §6. Frozen DINOv2-Base patch features  
 1014 at five resolutions (matching our Habitat chassis: blind, coarse, foveated, uniform, log-polar) feed  
 1015 a small LSTM trained on Memory Maze 9×9 offline trajectories with a next-feature prediction  
 1016 objective; we then probe agent\_pos from the LSTM hidden state (Ridge linear and 4-layer MLP  
 1017 probes, frame-level 80/20 split, eval window steps 250–500 of  $T=500$  trajectories).